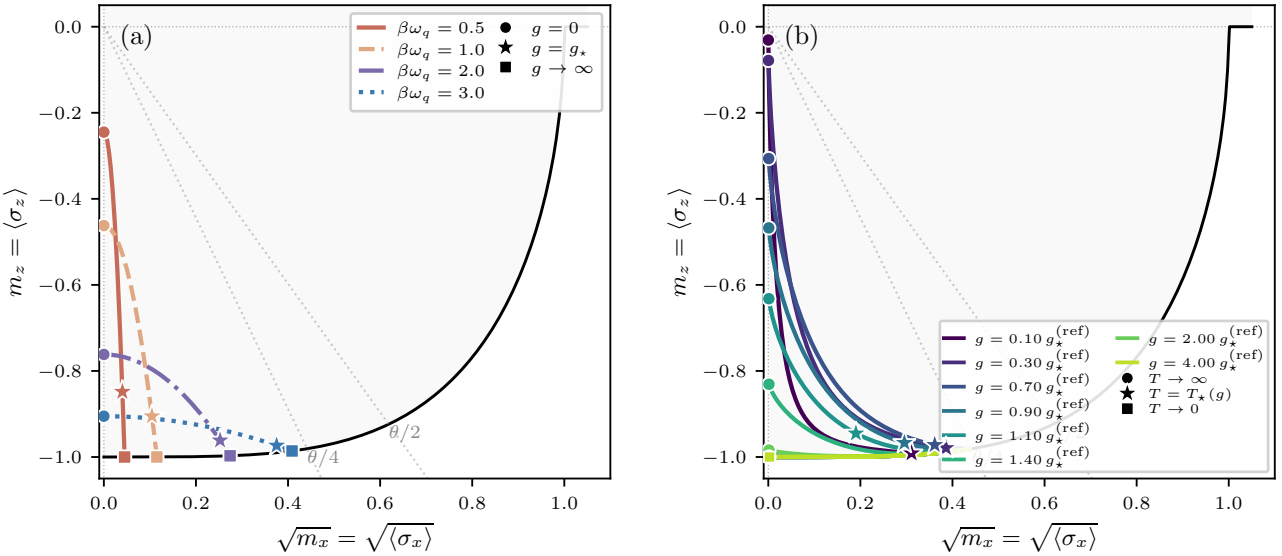
(a)
$\beta\omega_q = 0.5$
$\beta\omega_q = 1.0$
$\beta\omega_q = 2.0$
$\beta\omega_q = 3.0$
$g = 0$
$g = g_\star$
$g \to \infty$
$\theta/2$
$\theta/4$
$m_z = \langle\sigma_z\rangle$
$\sqrt{m_x} = \sqrt{\langle\sigma_x\rangle}$
(b)
$g = 0.10\, g_\star^{(\mathrm{ref})}$
$g = 0.30\, g_\star^{(\mathrm{ref})}$
$g = 0.70\, g_\star^{(\mathrm{ref})}$
$g = 0.90\, g_\star^{(\mathrm{ref})}$
$g = 1.10\, g_\star^{(\mathrm{ref})}$
$g = 1.40\, g_\star^{(\mathrm{ref})}$
$g = 2.00\, g_\star^{(\mathrm{ref})}$
$g = 4.00\, g_\star^{(\mathrm{ref})}$
$T \to \infty$
$T = T_\star(g)$
$T \to 0$
$m_z = \langle\sigma_z\rangle$
$\sqrt{m_x} = \sqrt{\langle\sigma_x\rangle}$